

CO2: Dry Run Evaluation — Answers

Course: Artificial Intelligence (CSA17) | Assessment Tool 4

Q1. Breadth-First Search (BFS)

Graph: A is the root, with children B and C; B has children D and E; C has children F and G.

BFS Trace Table (starting from A)

Iteration	Queue (before expansion)	Visited Node
1	A	A
2	B, C	B
3	C, D, E	C
4	D, E, F, G	D
5	E, F, G	E
6	F, G	F
7	G	G

Answers

1. **BFS traversal order:** $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$
2. **Queue contents after each iteration:** After iter 1: [B, C]; after iter 2: [C, D, E]; after iter 3: [D, E, F, G]; after iter 4: [E, F, G]; after iter 5: [F, G]; after iter 6: [G]; after iter 7: [] (empty)

Q2. Depth-First Search (DFS)

Using the same graph, starting from A (children explored left-to-right, so the left child is visited first).

DFS Trace Table

Iteration	Stack (before expansion)	Visited Node
1	A	A
2	C, B	B
3	C, E, D	D
4	C, E	E
5	C	C
6	G, F	F
7	G	G

DFS traversal order: $A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow G$

Note: B is expanded before C because it is pushed last / popped first (LIFO order). After B's subtree (D, E) is fully explored, the search backtracks to C.

Q3. 8-Puzzle — BFS Search

Initial State: 1 3 6 / 5 0 2 / 4 7 8 **Goal State:** 1 2 3 / 4 5 6 / 7 8 0

An actual breadth-first search was run over this puzzle (blank tile "0" moves Up/Down/Left/Right) to obtain accurate figures rather than estimates.

BFS Trace Table (first 5 iterations)

Iter	Current State Expanded	Queue Before (size)	Queue After (size)
1	1 3 6 / 5 0 2 / 4 7 8 (initial)	1	4
2	1 0 6 / 5 3 2 / 4 7 8	4	5
3	1 3 6 / 5 7 2 / 4 0 8	5	6
4	1 3 6 / 0 5 2 / 4 7 8	6	7
5	1 3 6 / 5 2 0 / 4 7 8	7	8

Because the blank tile typically has 2-4 possible moves per state, the queue grows quickly. BFS must fully expand every state at one depth level before moving to the next, so reaching the goal (which sits 8 moves deep) actually requires expanding hundreds of states, not just the 5 rows shown above. The table shows the correct start of the trace; a full trace continues in the same pattern for many more iterations.

Answers

1. Which node is expanded in each iteration? Each iteration expands the state at the front of the queue (standard FIFO order) — see the "Current State Expanded" column above; this continues level-by-level until the goal state is dequeued.

2. How many states are generated in total? Running the search fully: 838 successor states were generated (including revisits before being filtered), and 500 unique states were visited/expanded by the time the goal was reached.

3. Complete solution path (shortest path found, 8 moves)

Step	State	Move
0	1 3 6 / 5 0 2 / 4 7 8	(initial)
1	1 3 6 / 5 2 0 / 4 7 8	Right
2	1 3 0 / 5 2 6 / 4 7 8	Up
3	1 0 3 / 5 2 6 / 4 7 8	Left
4	1 2 3 / 5 0 6 / 4 7 8	Down
5	1 2 3 / 0 5 6 / 4 7 8	Left
6	1 2 3 / 4 5 6 / 0 7 8	Down

Step	State	Move
7	1 2 3 / 4 5 6 / 7 0 8	Right
8	1 2 3 / 4 5 6 / 7 8 0	Right

4. Why does BFS always find the shortest solution in an unweighted 8-puzzle? BFS explores the search space level by level — it fully expands all states at depth d before expanding any state at depth $d+1$. Since every move (tile slide) in the 8-puzzle has the same cost (1 step), the first time the goal state is dequeued, it is guaranteed to be at the shallowest possible depth — i.e. reached via the fewest possible moves. This makes BFS both complete and optimal for unweighted search spaces like this one.